

Solutions to Mock JEE Main - 19 | JEE-2022

PHYSICS

SECTION-1

1.(C) $X_L = \frac{100}{8}, R = \frac{100}{10} = 10\Omega; L \times 100\pi = \frac{100}{8} \text{ or } L = \frac{1}{8\pi} H$

$$Z = \sqrt{\left(\frac{1}{8\pi} \times 2\pi \times 40\right)^2 + 10^2} = 10\sqrt{2}$$

$$I = \frac{E}{Z} = \frac{100}{10\sqrt{2}} = \frac{10}{\sqrt{2}} = 5\sqrt{2} \text{ A}$$

Hence (C) is correct

- 2.(A) The path difference introduced by the plate of thickness t and refractive index μ is given by $\Delta = (\mu - 1)t$

Optical path difference (extra) shifts the fringe pattern by

$$y = \frac{(\mu - 1)\beta t}{\lambda} \quad (\text{where } \beta \text{ is fringe width})$$

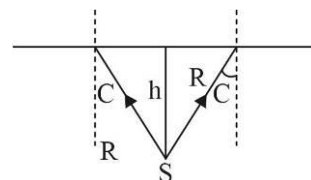
For first plate, $y = \frac{(\mu_1 - 1)\beta t}{\lambda}$

For second plate, $\frac{3}{2}y = \frac{(\mu_2 - 1)t\beta}{\lambda} \quad \therefore \frac{(\mu_2 - 1)}{(\mu_1 - 1)} = \frac{3}{2} \Rightarrow \frac{\mu_2 - 1}{1.5 - 1} = \frac{3}{2} \Rightarrow \mu_2 = 1.75$

3.(C) Fraction = $\frac{2\pi R^2(1 - \cos\theta)}{4\pi R^2}$ (where θ is $\sin^{-1}(3/4)$)

$$= \frac{1 - \cos\theta}{2} = \frac{1}{2} \left\{ 1 - \sqrt{1 - \left(\frac{9}{16}\right)} \right\} = 0.17$$

$$\therefore \text{Percentage} = 17\%$$



- 4.(C) The gain in K.E. of a charged particle after moving through a potential difference of V is given as eV , that is also equal to $\frac{1}{2}mv^2$, where v is the velocity of the charge particle. Disregarding the relativistic effect,

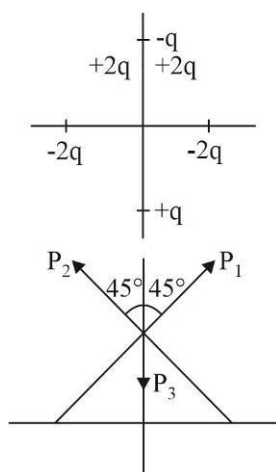
$$\frac{1}{2}mv^2 = qV \Rightarrow v = \sqrt{\frac{2qV}{m}}$$

$$\therefore mv = \sqrt{2mqV} \Rightarrow \text{De-broglie's wavelength} = \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}}$$

$$\therefore \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha V_\alpha}{m_p q_p V_p}}$$

Putting $V_\alpha = V_p$, $\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{(4)(2)}{(1)(1)}} = 2\sqrt{2}$

5.(A)



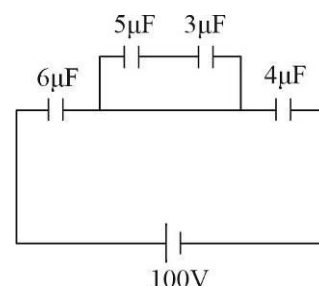
$$P_1 = P_2 = (2q)(\sqrt{2}a)$$

$$P_3 = q(2a) \quad \therefore P_{net} = (2P_1 \cos 45^\circ - P_3)\hat{j} = 2qa\hat{j}$$

6.(C) 6Ω , 3Ω , 8Ω , 4Ω & 10Ω resistors make a balanced wheat-stone bridge. Therefore 10Ω resistance can be ignored.

$$R_{AB} = 7 + \frac{9 \times 12}{9 + 12} = \frac{85}{7} \Omega$$

7.(D) Three capacitors of $2\mu F$ capacity each are in parallel. So, their combined capacity is $6\mu F$. They are in series with the capacitor of capacity $4\mu F$. But the remaining two capacitors i.e., $5\mu F$ & $3\mu F$ are short circuited. Therefore, charge on both of them is zero. The simple circuit is shown below.



8.(B) Force on semicircular wire = force on section PQ

$$\therefore F = i(l \times B)$$

$$\text{Here, } l = PQ = (2 \cos 45^\circ)\hat{i} + (2 \sin 45^\circ)\hat{j}$$

$$= (\sqrt{2}\hat{i} + \sqrt{2}\hat{j})m$$

$$\therefore F = (1) \left[(\sqrt{2}\hat{i} + \sqrt{2}\hat{j}) \times (3\hat{i} + 4\hat{j} + \hat{k}) \right] = \sqrt{2}(\hat{i} - \hat{j} + \hat{k})$$

$$9.(B) \quad L = \frac{L_1 L_2}{L_1 + L_2} = 2H; \quad i_0 = \frac{E}{R} = 1A$$

$$\text{Now, } \frac{\frac{1}{2} L i_0^2}{i_0^2 R t} = \frac{L}{2 R t} = \frac{2}{2 \times 10 \times 10} = \frac{1}{100}$$

$$10.(B) \quad R = 2\sqrt{h(H-h)}$$

R will be same for two heights h_1 & h_2 , such that $h_1 + h_2 = H$

$$30 + h_2 = 50$$

$$h_2 = 20cm$$

11.(A) For liquid

$$\rho_b Vg = \rho_L Vg ; \quad \rho_b = \rho_L$$

For liquid 2

$$\rho_b Vg = 4\rho_L V'g \quad [V' \text{ is submerged volume}]$$

$$\frac{V'}{V} = \frac{\rho_b}{4\rho_L} \quad (\text{also } \rho_b = \rho_L)$$

$$\frac{V'}{V} = \frac{1}{4}; \quad \frac{V'}{V} \times 100 = 25\%$$

12.(D) Case-I

$$50 S_p (80 - T) = 100 S_Q (T - 40)$$

Also $T = 60$

$$50 S_p (80 - 60) = 100 S_Q (60 - 40)$$

$$S_p = 2S_Q \quad \dots(i)$$

Case-II

$$25 S_p (100 - T) = 100 S_Q (T - 25) \quad \dots(ii)$$

From (i) and (ii)

$$50 S_Q (100 - T) = 100 S_Q (T - 25)$$

$$100 - T = 2T - 50$$

$$3T = 150 \Rightarrow T = 50^\circ C$$

13.(B) $(60 - 52) = K \left(\frac{60 + 52}{2} - 16 \right) \times 10$

$$8 = K(40) \times 10$$

$$\frac{1}{5} = 10K \quad \dots(i)$$

Now,

$$(52 - T) = K \left(\frac{52 + T}{2} - 16 \right) \times 10 \quad \dots(ii)$$

From (i) and (ii)

$$52 - T = \frac{1}{5} \frac{(52 + T - 32)}{2}$$

$$520 - 10T = 20 + T; \quad 11T = 500 \Rightarrow T \approx 45^\circ C$$

14.(B) Energy stored per unit volume (E) = $\frac{1}{2} \frac{(\text{stress})^2}{y}$

$$= \frac{1}{2} \left(\frac{F}{A} \right)^2 \frac{1}{y}; \quad E_1 = \frac{1}{2} \left(\frac{\frac{w}{\pi \frac{d_1^2}{4}}}{y} \right)^2 \frac{1}{y}$$

$$E_2 = \frac{1}{2} \left(\frac{2w}{\pi \frac{d_2^2}{4}} \right)^2 \frac{1}{y}; \quad \frac{E_1}{E_2} = \left(\frac{d_2^2}{2d_1^2} \right)^2$$

$$\frac{1}{64} = \left(\frac{d_2^2}{2d_1^2} \right)^2; \quad \frac{d_2^2}{d_1^2} = \frac{1}{4} \Rightarrow \frac{d_1}{d_2} = \frac{2}{1}$$

15.(B) $\eta = \frac{W}{Q_{in}} = \frac{W}{W + Q_{out}} = \frac{900}{1920}$

$\eta_{carnot} = 1.3\eta = 1.3 \times \frac{900}{1920} = 0.6$

$\Rightarrow \frac{T_C}{T_H} = 1 - 0.6 \Rightarrow T_C \cong 280 K$

16.(A) $0.6 \times (T - 300) = 0.09(500 - T)$

$20T - 6000 = 1500 - 3T$

$T = \frac{7500}{23} = 326 K = 53^\circ C$

17.(B) As lift is moving up and slowing down

$\Rightarrow$ acceleration of lift $= \bar{a} = 5 m/s^2 \downarrow$

$\Rightarrow$ Effective g in lift $= g - 5 = 5 m/s^2 = \frac{g}{2}$

$\Rightarrow T_{new} = 2 = \sqrt{\frac{l}{g/2}} = \sqrt{2} \times T_0 = 2\sqrt{2}s = 2.82s$

18.(D) At $x = -3cm$ displacement from mean position is 5 cm

$\Rightarrow KE = \frac{1}{2} K(A^2 - x^2) = \frac{1}{2} K(100 - 25) = \frac{75K}{2}$

$\Rightarrow PE = \frac{1}{2} Kx^2 = \frac{25K}{2} \Rightarrow \frac{KE}{PE} = 3$

19.(B) $c = \frac{\omega}{k} = 45 m/s = \sqrt{\frac{T}{\mu}}$

$T = \frac{0.12 \times 10^{-3}}{10^{-2} m} kg \times 45 \times 45 = 24.3 N$

20.(D) For 1st resonance, $\frac{\lambda}{4} = l + e = l + 0.3d$

$= 15 + 0.3 \times 10 = 18 cm$

$\Rightarrow \lambda = 72 cm$

$f = \frac{c}{\lambda} = \frac{330}{0.72} \approx 460 Hz$

SECTION-2

21.(2) $r = c$, each surface

$$A = r + r = 60^\circ$$

$$r = 30^\circ; \quad \mu = \frac{\sin 90^\circ}{\sin 30^\circ} = 2$$

22.(41) For K_α X-ray line,

$$\frac{1}{\lambda_\alpha} = R(Z-1)^2 \left[\frac{1}{1^2} - \frac{1}{2^2} \right] = R(Z-1)^2 \left[1 - \frac{1}{4} \right]$$

$$\Rightarrow \frac{1}{\lambda_\alpha} = \frac{3}{4} R(Z-1)^2 \quad \dots(i)$$

With reference to given data,

$$\lambda_\alpha = 0.76 \text{ \AA} = 0.76 \times 10^{-10} \text{ m}; \quad R = 1.097 \times 10^7 \text{ m}^{-1}$$

Putting these values in equation (i)

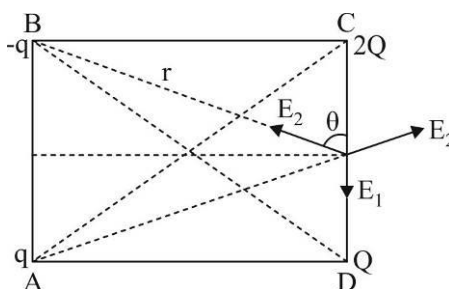
$$(Z-1)^2 = \frac{4}{3} \frac{1}{0.76 \times 10^{-10} \times 1.097 \times 10^7} \cong 1600 \quad \Rightarrow \quad Z-1 = 40 \Rightarrow Z = 41$$

23.(2) $E_1 = 2E_2 \cos \theta$

$$\therefore \frac{KQ}{(a/2)^2} = 2 \left[\frac{Kq}{r^2} \cdot \frac{a/2}{r} \right]$$

$$\therefore \frac{q}{Q} = 4 \left(\frac{r}{a} \right)^3$$

$$= 4 \left(\sqrt{\frac{5}{4}} \right)^3 = \frac{5\sqrt{5}}{2}$$



24.(9) Potential difference between A and B = 6 volts. The condensers $2\mu F$ & $5\mu F$ are in parallel. Their effective capacitance, $C = 2 + 5 = 7\mu F$.

The capacitance between A and B is given by

$$C' = \frac{7 \times 3}{7 + 3} = \frac{21}{10} \mu F$$

$$\text{Total charge } Q = CV = \frac{21}{10} \times 6 = \frac{63}{5} \mu C$$

Total potential difference across $3\mu F$ is

$$V_1 = \frac{Q}{3} = \frac{63}{5} \times \frac{1}{3} = \frac{21}{5} V$$

Hence, the common potential difference across the condensers in parallel is

$$V_2 = 6 - \frac{21}{5} = \frac{9}{5} V$$

So, the charge on $5\mu F$ condenser is

$$Q = 5 \times V_2 = 5 \times \frac{9}{5} = 9 \mu C$$

25.(2) Flux passing from the small loop,

$$\phi = BS = \left(\frac{\mu_0 i}{2R} \right) (a^2)$$

$$e = \frac{d\phi}{dt} = \left(\frac{\mu_0 a^2}{2R} \right) \frac{di}{dt} \text{ in volts}$$

$$= \left(\frac{\mu_0 a^2}{2R} \right) \left(\frac{i_0}{\tau} \right) \times 10^6 \text{ (in microvolts)}$$

$$= (4\pi \times 10^{-7})(10^{-2})^2 \times \frac{2}{\pi} \times 100 \times 10^6 \times \frac{1}{2 \times 0.1 \times 2 \times 10^{-2}} = 2\mu V$$

26.(4) $V_2 = \frac{V_1}{8} \Rightarrow \frac{4}{3}\pi r_2^3 = \frac{1}{8} \left(\frac{4}{3}\pi r_1^3 \right)$

$$r_2 = \frac{r_1}{2}$$

Now, $\frac{P_2}{P_1} = \frac{\sigma A_2 T_2^4}{\sigma A_1 T_1^4}$

$$\frac{P_2}{P_1} = \frac{(\sigma)(4\pi r_2^2)(2T)^4}{(\sigma)(4\pi r_1^2)(T)^4}; \quad \frac{P_2}{P_1} = \frac{1}{4}(2)^4 \Rightarrow \frac{P_2}{P_1} = 4$$

27.(125) $\frac{4}{3}\pi R^3 = n \times \frac{4}{3}\pi r^3$ (R is radius of big drop ; r is radius of small drop)

$$r = \frac{R}{(n)^{1/3}}$$

Now, $\frac{5U}{U} = \frac{T(4\pi r^2 \times n)}{T(4\pi R^2)}$

$$5 = \left(\frac{R^2}{(n)^{2/3}} \times n \right) \frac{1}{R^2} \Rightarrow n^{1/3} = 5 \Rightarrow n = 125$$

28.(180) Process AC is isochoric $\Rightarrow Q_{AC} = \Delta U_{AC} = 70J$

$$\Delta U_{AB} = \Delta U_{AC} + \Delta U_{CB} = 70 + 50J = 120J$$

$$Q_{ADB} = W_{ADB} + \Delta U_{AB} = 60 + 120J = 180J$$

29.(6) $A = \frac{A_0}{5} = A_0 e^{\frac{-bt}{2m}} \Rightarrow \frac{bt}{2m} = \ln 5 \Rightarrow t = \frac{2m \ln 5}{b}$

$$t = \frac{2 \times 200 \times 0.7 \times 2.3}{10} = 64.4 \approx 2^6$$

30.(8) $\Delta\lambda = \lambda \times \frac{V}{c} = 600 \times 10^{-9} \times \frac{410 \times 10^3}{3 \times 10^8} = 8 \times 10^{-10} m$

Chemistry

SECTION-1

- 1.(C) It resembles with FCC where tetrahedral and octahedral voids are occupied by the atoms

$$\sqrt{2}a = 4r; \sqrt{2}a = 4 \times 1 \Rightarrow a = 2\sqrt{2}$$

No. of unit cells formed by n atoms present in packing $\frac{n}{4}$ (fcc)

$$\text{Volume of crystal} = \frac{n}{4} a^3 = 4\sqrt{2}n$$

Volume of atoms

$$= \frac{n}{4} \left[4 \times \frac{4}{3} \pi (1)^3 + 4 \times \frac{4}{3} \pi (0.414)^3 + 4 \times 2 \times \frac{4}{3} \pi (0.225)^3 \right] = 4.58n$$

$$\% \text{ space occupied} = \frac{4.58n}{4\sqrt{2}n} \times 100 = 81\%$$

$$\% \text{ void} = 100 - 81 = 19\%$$

- 2.(A) $\text{AgI} + e^- \rightleftharpoons \text{Ag} + \text{I}^-$

$$\Delta G^\circ = -nFE^\circ = +0.164F$$

$$\text{Ag} \rightleftharpoons \text{Ag}^+ + e^-; \Delta G^\circ = -nFE^\circ = +0.8F$$

By adding these two equations, we get

$$\text{AgI} \rightleftharpoons \text{Ag}^+ + \text{I}^-; \Delta G^\circ = 0.164F + 0.8F = 0.964F$$

$$\log K_{sp} = \frac{-\Delta G^\circ}{2.303RT}$$

$$\log K_{sp} = \frac{-0.964 \times 96500}{2.303 \times 8.314 \times 298}; \quad K_{sp} = 5 \times 10^{-17}$$

- 3.(A) Let the half lives of A and B are 2a and 3a seconds respectively

For A : at $T = 3 \times 2a = 6a$ seconds

$$\ln \frac{1}{(n_A)_t} = \left[\frac{\ln 2}{2a} \right] 6a \quad \dots(i)$$

$$\ln \frac{1}{(n_A)_t} = \ln 2^3; \quad (n_A)_t = \frac{1}{8}$$

(Given initially $n_A = n_B = 1$)

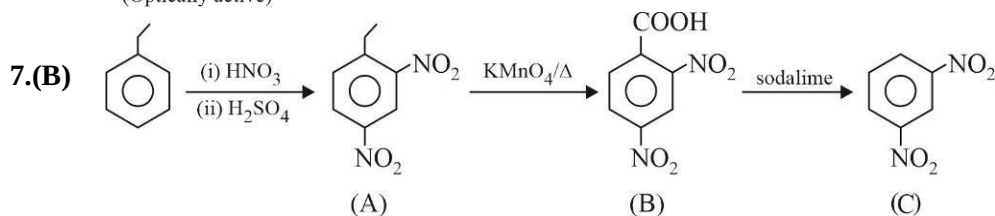
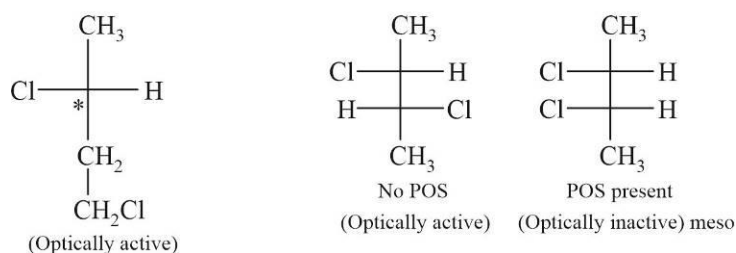
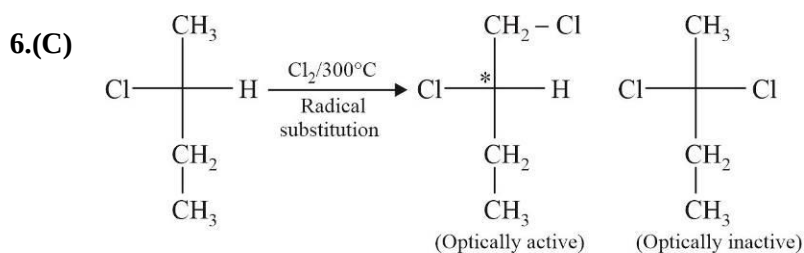
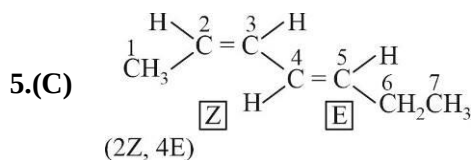
For B : at $T = 6a$ seconds

$$\ln \frac{1}{(n_B)_t} = \left[\frac{\ln 2}{3a} \right] 6a \quad \dots(ii)$$

$$(n_B)_t = 1/4$$

$$\text{From (i) and (ii), } \frac{(n_A)_t}{(n_B)_t} = \frac{1}{2}$$

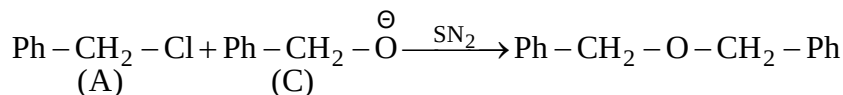
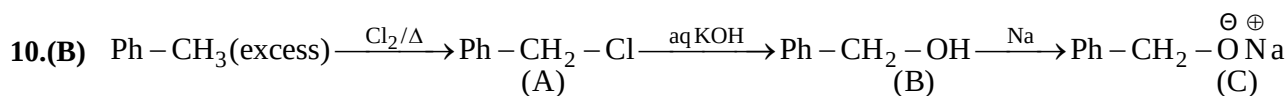
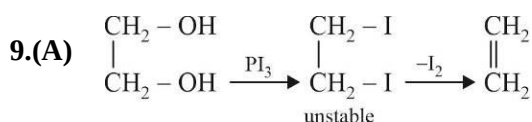
4. (A) Informative



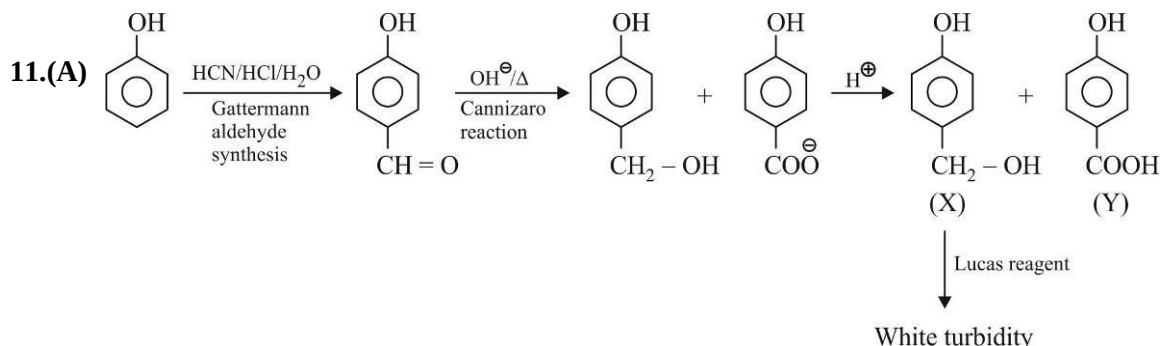
8.(A) Substrate is tertiary halide so reaction is SN_1

$$\text{Thus, } \frac{dx}{dt} = K [\text{substrate}]$$

Rate of SN_1 reaction is independent of concentration of alkali.

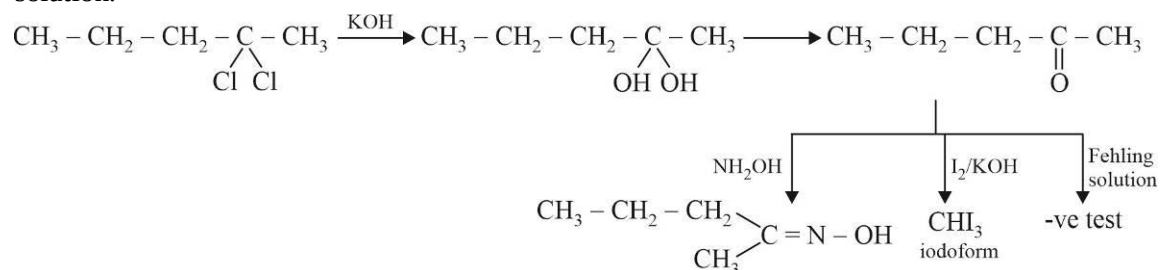


williamson synthesis

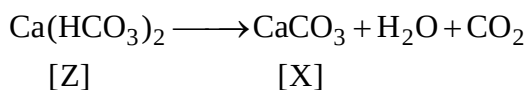
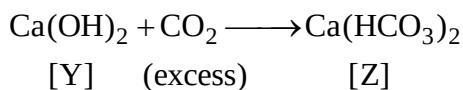
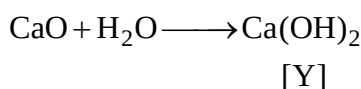


Both are phenolic so that (X) and (Y) give litmus test.

- 12.(A) $C_5H_{10}O$ must be methyl ketone because it gives iodoform. Also ketone does not answer Fehling's solution.



- 13.(A) $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$
[X]



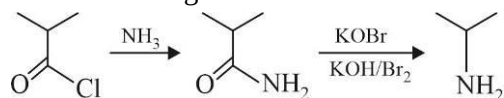
- 14.(A) When acidified $KMnO_4$ reacts with oxalic acid then reaction is slow in the beginning but becomes very rapid because Mn^{+2} acts as autocatalyst.

- 15.(C) Correct order of bond dissociation energy
 $Cl_2 > Br_2 > F_2 > I_2$

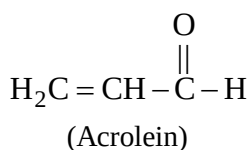
- 16.(D)

Complex	Number of unpaired e^-
$[MnBr_4]^{2-}$	5
$[Fe(CN)_6]^{4-}$	0
$[CoBr_4]^{-2}$	3

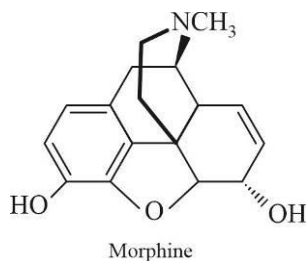
- 17.(C) In Hoffman degradation one carbon decreases



- 18.(D) $C_3H_4O \longrightarrow$ unsaturated aldehyde

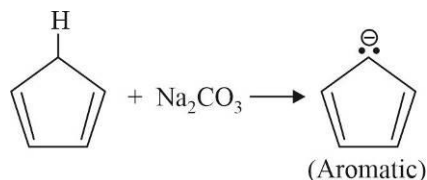


19.(A)



Represents the correct structure of morphine

20.(C)



SECTION-2

21.(5) $1230 = P_{\text{gas}} + P_{\text{water vapour}}$

$$P_{\text{gas}} = 1200 \text{ torr}$$

As volume of container is doubled

(i) V.P. of water remain same (i.e. 30 torr)

(ii) Gas pressure changes (i.e., becomes half, 600 torr)

Total pressure = 630 torr

22.(9) No. of Faradays

$$= \frac{8 \times 10^{-3} \times 60}{96000} = \frac{480 \times 10^{-3}}{96000} = 5 \times 10^{-6}$$

Equivalents of NaOH = 5×10^{-6}

Equivalents of Cl_2 released = 5×10^{-6}

Equivalents of H_2 released = 5×10^{-6}

Normality of NaOH = $\frac{5 \times 10^{-6}}{500} \times 1000 = 10 \times 10^{-6} = 10^{-5}$

$[\text{OH}^-] = 10^{-5} \text{ M}$, pOH = 5, pH = 9

23.(5) $A_n \rightarrow nA$

$t = 0 \quad a_0 \quad -$

$t = t \quad a_0 - \frac{a_0}{6} \quad \frac{na_0}{6}$

In graph, at time t , $[A_n]$ & $[A]$ are same

$a_0 - \frac{a_0}{6} = \frac{na_0}{6}$, here, $n = 5$

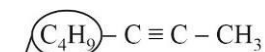
24.(6) $C_n H_{2n-2} = 96$

$12n + 2n - 2 = 96$

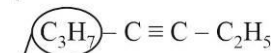
$14n = 98$

$n = 7$

$$\text{m.f.} = \text{C}_7\text{H}_{12}$$



4



2

Number of possible alkyne = 4 + 2 = 6

25.(6) Reagent (2), (3), (4), (5), (6), (7) reacts with the given compound.

26.(4) Oxide ores = 4

Cu_2O : **Cuprite**

PbS : **Galena**

SnO_2 : **Cassiterite**

Fe_3O_4 : **Magnetite**

Al_2O_3 : **Bauxite**

$\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$: **Carnallite**

ZnCO_3 : **Calamine**

27.(5) Vitamin B and C are water soluble and Vitamin A, D, E and K are fat soluble.

28.(3) 1 degree aromatic amines undergo diazotisation

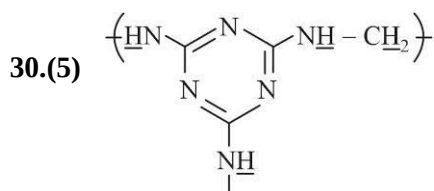
Hence, III, V, VI undergo diazotisation

29.(1) Acidic $\rightarrow \text{CO}_2, \text{SiO}_2, \text{GeO}_2, \text{GeO}$ (a = 4)

Neutral $\rightarrow \text{CO}$ (b = 1)

Amphoteric $\rightarrow \text{SnO}, \text{PbO}, \text{SnO}_2, \text{PbO}_2$ (c = 4)

So, $a + b - c = 4 + 1 - 4 = 1$



Melamine polymer

No. of hydrogen atoms = 5

Mathematics

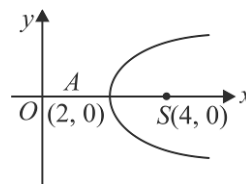
SECTION-1

1.(D) We have, vertex $A(2, 0)$ and focus $S(4, 0)$.

$\therefore$ Equation of parabola will be $y^2 = 4 \times 2(x - 2)$

$$\Rightarrow y^2 = 8(x - 2)$$

Clearly, $(8, 6)$ will not lie on this parabola.



2.(D) We are given that

$$\therefore f(x) = \min(2 \sin x, 1 - \cos x, 1)$$

$$\int_0^\pi f(x) dx = \int_0^{\pi/2} (1 - \cos x) dx + \int_{\pi/2}^{5\pi/6} dx + \int_{5\pi/6}^\pi 2 \sin x dx$$

$$= \frac{\pi}{2} - 1 + \frac{5\pi}{6} - \frac{\pi}{2} - 2 \left[-1 + \frac{\sqrt{3}}{2} \right]$$

$$= 1 - \sqrt{3} + \frac{5\pi}{6}$$

3.(A) In 2×2 matrices total positions are 4 in each position there are 2 choices.

$$\therefore n(S) = 2^4 = 16$$

Matrices which are not singular are

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

$\therefore$ Total matrices which are singular = $16 - 6 = 10$

$$\therefore \text{Required probability} = \frac{10}{16} = \frac{5}{8}$$

4.(C) The given equations are

$$4^{\sin x} + 3^{\sec y} = 1 \quad \dots (i)$$

$$\text{and } 5(16)^{\sin x} - 2(3)^{\sec y} = 2 \quad \dots (ii)$$

$$\text{Let } \lambda = 4^{\sin x}$$

$$\mu = 3^{\sec y}$$

Then, equations (i) and (ii) reduced to

$$\lambda + \mu = 11$$

$$5\lambda^2 - 2\mu = 2 \Rightarrow 5\lambda^2 - 2(11 - \lambda) = 2$$

$$\Rightarrow 5\lambda^2 + 2\lambda - 24 = 20 \Rightarrow \lambda = \frac{-2 \pm \sqrt{484}}{10} \Rightarrow \lambda = 2, \frac{-12}{5}$$

$$\text{For } \Rightarrow 4^{\sin x} = 2 \text{ and } 3^{\sec y} = 9 \Rightarrow 2 \sin x = 1 \text{ and } \sec y = 2$$

$$\Rightarrow \sin x = \frac{1}{2} \text{ and } \cos y = \frac{1}{2} \text{ and } x = n\pi + (-1)^n \frac{\pi}{6} \text{ and } y = 2m\pi \pm \frac{\pi}{3}, m, n \in \mathbb{Z}$$

For $\lambda = -\frac{12}{5}$

$\Rightarrow (4)^{\sin x} = -\frac{12}{5}$ which is not possible $4^{\sin x} > 0$.

Thus, the solution of the given equations is given by

$$x = n\pi + (-1)^n \frac{\pi}{6}, y = 2m\pi \pm \frac{\pi}{3}, m, n \in \mathbb{Z}$$

5.(D) Let $f = (5\sqrt{2} - 7)^{19}$

$x - f = \text{an integer} \Rightarrow [x] + \{x\} - f = \text{an integer}$

$\Rightarrow \{x\} - f = \text{an integer, but } -1 < \{x\} - f < 1 \Rightarrow \{x\} = f$

So, $x\{x\} = x \cdot f = 1^{19} = 1$

6.(C) $\because y^2 = 4ax$; equation of normal at $(at^2, 2at)$ is $y + tx = 2at + at^3$

Here, equation of the normal chord at any point $(at^2, 2at)$ of the parabola is

$$y + tx = 2at + at^3 \quad \dots(i)$$

Equation of the chord with mid-point (x_1, y_1) is $T = S_1$

$$yy_1 - 2ax = y_1^2 - 2ax_1 \quad \dots(ii)$$

Since, equation (i) and (ii) are identical

$$\frac{1}{y_1} = \frac{t}{-2a} = \frac{2at + at^3}{y_1^2 - 2ax_1}$$

$$t = \frac{-2a}{y_1} \text{ and } \frac{y_1^2 - 2ax_1}{-2a} = \frac{2at + at^3}{t} = 2a + a\left(\frac{-2a}{y_1}\right)^2$$

Or $\frac{-y_1^2}{2a} + x_1 = 2a + \frac{4a^3}{y_1^2} \Rightarrow x_1 - 2a = \frac{y_1^2}{2a} + \frac{4a^3}{y_1^2}$

Hence, the locus of the middle point (x_1, y_1) is $x - 2a = \frac{y^2}{2a} + \frac{4a^3}{y^2}$

7.(B) Differentiating w.r.t. x , we get :

$$\frac{2\cos x - \sin x + \lambda}{\cos x + \sin x - 2} = \frac{d}{dx} (A \ln |\cos x + \sin x - 2| + Bx + C)$$

$$= A \frac{(\cos x - \sin x)}{\cos x + \sin x - 2} + B = \frac{A \cos x - A \sin x + B \cos x + B \sin x - 2B}{\cos x + \sin x - 2}$$

$$\therefore \frac{2\cos x - \sin x + \lambda}{\cos x + \sin x - 2} = \frac{(A+B)\cos x + (B-A)\sin x - 2B}{\cos x + \sin x - 2}$$

$$A+B=2, B-A=-1, \lambda=-2B \Rightarrow A=\frac{3}{2}, B=\frac{1}{2}, \lambda=-1$$

8.(B) Here equation of line passing through the points (x_1, y_1) and (x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

Given that the equation of line AB is

$$2y = x \quad \dots \text{(i)}$$

Let coordinates of A be (h, k)

$$\therefore AB = AC$$

$$\Rightarrow AB^2 = AC^2$$

$$(k-1)^2 + (h-2)^2 = (h-1)^2 + (k-2)^2$$

$$h^2 + k^2 - 2k - 4h + 1 + 4 = h^2 + k^2$$

$$-2h - 4k + 1 + 4 = 0$$

$$h = k$$

Also, A lies on equation (i)

$$\Rightarrow 2k = k \Rightarrow k = 0$$

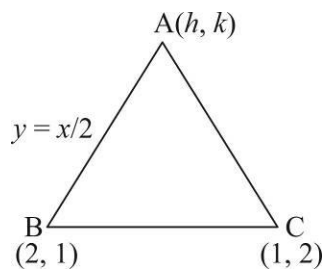
Simplifying above equation.

$$h = k = 0 \quad \therefore \text{Coordinates of A are } (0, 0)$$

Equation of line AC

$$\Rightarrow y - 0 = 2(x - 0)$$

$$y = 2x$$



9.(C) $y^2 = p(x) \Rightarrow 2y \frac{dy}{dx} = p'(x) \quad \dots \text{(i)}$

$$\Rightarrow \left(\frac{dy}{dx} \right)^2 + y \frac{d^2y}{dx^2} = \frac{p''(x)}{2}$$

$$\Rightarrow y \frac{d^2y}{dx^2} = \frac{p''(x)}{2} - \left(\frac{dy}{dx} \right)^2$$

Multiplying y^2 on both sides :

$$y^3 \frac{d^2y}{dx^2} = \frac{p(x)p''(x)}{2} - \left(y \frac{dy}{dx} \right)^2$$

$$= \frac{p(x)p''(x)}{2} - \frac{(p'(x))^2}{4}$$

$$= \frac{1}{4} \left[2p(x)p''(x) - (p'(x))^2 \right]$$

$$\Rightarrow 2y^3 \frac{d^2y}{dx^2} = \frac{1}{2} \left[2p(x)p''(x) - (p'(x))^2 \right]$$

On differentiating both sides

$$2 \frac{d}{dx} \left(y^3 \frac{d^2y}{dx^2} \right)$$

$$= \frac{1}{2} \left[2p'(x)p''(x) + 2p(x)p'''(x) - 2p'(x)p''(x) \right] = p(x)p'''(x)$$

10.(C) $[-\sin^2 x] = 0$ or -1 but $\sec^{-1}(0)$ is not defined. Hence, $y = \sec^{-1}[-\sin^2 x] = \pi$

Now, $\pi = \frac{16-x^2}{4} \Rightarrow x = \pm 2\sqrt{4-\pi}$

Hence, required area = $\int_{-2\sqrt{4-\pi}}^{2\sqrt{4-\pi}} \left(\frac{16-x^2}{4} - \pi \right) dx = \frac{8}{3}(4-\pi)^{3/2}$

11.(A) $\sin^4 x = \sin^2 x - \frac{1}{4}\sin^2 2x$

$\Rightarrow S_n = \left(\sin^2 x - \frac{1}{4}\sin^2 2x \right) + \left(\frac{1}{4}\sin^2 2x - \frac{1}{4^2}\sin^2 4x \right) + \dots + \left(\frac{1}{4^n}\sin^2 2^n x - \frac{1}{4^{n+1}}\sin^2 2^{n+1} x \right)$

$\Rightarrow S_n = \sin^2 x - \frac{1}{4^{n+1}}\sin^2 2^{n+1} x \Rightarrow \lim_{n \rightarrow \infty} S_n = \sin^2 x$

12.(C) Centre $(0, -f)$

Equation of circle $x^2 + y^2 + 2fy + c = 0$

$(\log_e a_r)^2 + \frac{1}{(\log_e a_r)^2} + 2f \left(\frac{1}{\log_e a_r} \right) + c = 0$

$(\log_e a_r)^2 + c(\log a_r)^2 + 2f(\log a_r) + 1 = 0$

$\log a_1, \log a_2, \log a_3, \log a_4$ are roots of equation

$\therefore \log(a_1 a_2 a_3 a_4) = 0$

$a_1 a_2 a_3 a_4 = 1$

$\frac{a_1 + a_2 + a_3 + a_4}{4} \geq (1)^{1/4}$

$\therefore a_1 + a_2 + a_3 + a_4 \geq 4$

13.(A) Now $AH \perp BC$

Now D^r of $AH = (\alpha - 6, \beta, \gamma)$

D^r of $BC = (0, 4, -2)$

Now for $\perp$ lines,

$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

$\therefore 0(\alpha - 6) + \beta(4) - 2\gamma = 0$

$\therefore 4\beta - 2\gamma = 0$

$2\beta = \gamma \quad \dots \text{(i)}$

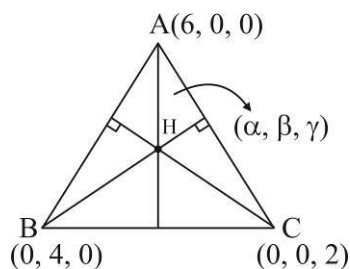
Similarly

$-6\alpha + 2\gamma = 0 \quad \dots \text{(ii)}$

$-6\alpha + 4\beta = 0 \quad \dots \text{(iii)}$

$\therefore 2\beta = \gamma; \frac{2\beta}{3} = \alpha$

Now H lies on



$$\begin{aligned}
 2x + 3y + 3z &= 12 \\
 \therefore 2\alpha + 3\beta + 6\gamma &= 12 \\
 2\left(\frac{2\beta}{3}\right) + 3\beta + 6(2\beta) &= 12 \\
 4\beta + 9\beta + 36\beta &= 36 \\
 49\beta &= 36 \\
 \beta &= \frac{36}{49} \\
 \text{Now, } \alpha &= \frac{24}{49} \text{ and } \gamma = \frac{72}{49} \\
 (\alpha, \beta, \gamma) &= \left(\frac{24}{49}, \frac{36}{49}, \frac{72}{49}\right)
 \end{aligned}$$

14.(D) Taking first two

$$\begin{aligned}
 \frac{\tan(\pi/4 + \alpha)}{\tan(\pi/4 + \beta)} &= \frac{5}{3} \\
 \therefore \frac{\cos(\alpha + \beta)}{\sin(\alpha - \beta)} &= \frac{8}{2} = 4 \\
 \therefore \cos(\alpha + \beta) &= 4 \sin(\alpha - \beta) \quad \dots \text{(i)}
 \end{aligned}$$

Similarly

$$\begin{aligned}
 \cos(\beta + \gamma) &= 5 \sin(\beta - \gamma) \quad \dots \text{(ii)} \\
 3 \cos(\gamma + \alpha) &= -7 \sin(\gamma - \alpha) \quad \dots \text{(iii)}
 \end{aligned}$$

After marking $\sin^{-2}$ terms

$$(i) + (ii) - (iii)$$

$$\text{Then } 12 \sin^2(\alpha - \beta) + 15 \sin^2(\beta - \gamma) - 7 \sin^2(\gamma - \alpha) = 0$$

$$15.(C) \quad y \left(xy - (x^2 - y^2)^2 \right) dx = \left\{ y^3 - x(x^2 - y^2)^2 \right\} dy$$

$$\Rightarrow xy^2 dx - y(x^2 - y^2)^2 dx = y^3 dy - x(x^2 - y^2)^2 dy$$

$$\Rightarrow y^2(x dx - y dy) = (x^2 - y^2)^2(y dx - x dy) \Rightarrow x dx - y dy = (x^2 - y^2)^2 \frac{(y dx - x dy)}{y^2}$$

$$\frac{2(y dx - x dy)}{(x^2 - y^2)^2} = \frac{2(y dx - x dy)}{y^2}$$

$$\int \frac{d(x^2 - y^2)}{(x^2 - y^2)^2} = 2 \int d\left(\frac{x}{y}\right)$$

$$\frac{-1}{x^2 - y^2} = \frac{2x}{y} + C \quad \therefore \frac{2x}{y} + \frac{1}{x^2 - y^2} = C$$

16.(A) Here direction ratios of the line AB are 1, 2, 2.

So direction cosine are

$$\frac{1}{\sqrt{1^2+2^2+2^2}}, \frac{2}{\sqrt{1^2+2^2+2^2}}, \frac{2}{\sqrt{1^2+2^2+2^2}}$$

i.e., $\frac{1}{3}, \frac{2}{3}, \frac{2}{3}$

Projection of AP on the line AB i.e.,

$$\begin{aligned} AN &= 1(x-a) + m(y-b) + n(z-c) \\ &= \frac{1}{3}(0-1) + \frac{2}{3}(-1+3) + \frac{2}{3}(3-2) \\ &= \frac{5}{3} \end{aligned}$$

Also length of AP = $\sqrt{6}$ units

$\therefore$ The perpendicular distance of point $P(0, -1, 3)$ from the straight line passing through

$$A(1, -3, 2) \quad \text{and} \quad B(2, -1, 4) = \sqrt{6 - \left(\frac{5}{3}\right)^2} = \frac{\sqrt{29}}{3} \text{ units.}$$

17.(B) $f'(x) = -\sin 4x$

$$\because f'(x) > 0 \Rightarrow -\sin 4x > 0 \Rightarrow \sin 4x < 0$$

$$\text{or} \quad (2n+1)\pi < 4x < (2n+2)\pi$$

$$\Rightarrow (2n+1)\frac{\pi}{4} < x < \frac{(n+1)\pi}{2}$$

$$\text{For } n=0; \frac{\pi}{4} < x < \frac{\pi}{2}$$

$$\text{Now, } \frac{4\pi}{8} > \frac{3\pi}{8}$$

$$\therefore f(x) \text{ is increasing in } \left(\frac{\pi}{4}, \frac{3\pi}{8}\right)$$

18.(A) We have given that $f(x) = \cos x$

$$\because \frac{-\pi}{3} < x \leq \frac{\pi}{6} \quad \because \cos x \text{ is decreasing function}$$

$$\frac{-\pi}{3} < x < 0 \text{ and } 0 \leq x \leq \frac{\pi}{6}$$

$$\cos \frac{\pi}{3} \leq \cos x < \cos 0 \text{ or } \cos \frac{\pi}{6} \leq \cos x \leq \cos 0$$

$$\frac{\sqrt{3}}{2} < \cos x \leq 1 \text{ or } \frac{1}{2} \leq \cos x \leq 1$$

$$\therefore \frac{1}{2} < \cos x \leq 1$$

$$a = \frac{1}{2}, b = 1; \quad a + b = \frac{1}{2} + 1 = \frac{3}{2}$$

$$19.(A) \quad I = \int_1^e (\sin x) \left(\frac{1}{x} \right) dx + \int_1^e (x \ln x - x) \sin x dx.$$

Applying integration by parts, we get

$$I = \sin x \ln x \Big|_1^e - \int_1^e \cos x \ln x dx - (x \ln x - x) \cos x \Big|_1^e + \int_1^e \ln x \cdot \cos x dx = \sin e - \cos 1$$

20.(C) The given differential equation can be written as $\frac{dx}{dy} + \frac{1}{y^2} x = \frac{1}{y^3}$ which is a linear equation with I.F.

$= e^{\int (1/y^2) dy} = e^{-1/y}$. Multiplying with the integrating factor

$$\frac{d}{dx} (x e^{-1/y}) = \frac{1}{y^3} e^{-1/y} \Rightarrow x e^{-1/y} = \int \frac{1}{y^3} e^{-1/y} dy + C = \int -u e^u du + C \quad (u = -1/y)$$

SECTION-2

21.(4) Consider the four points A, B, C, D with position vectors

$$OA = a, OB = b, OC = c \text{ and } OD = d$$

It is given that $|AB \times CD + BC \times AD + CA \times BD|$

$= k \cdot \text{area of } \Delta ABC$

$$\Rightarrow 2|a \times b + b \times c + c \times a| = k \cdot \text{area of } \Delta ABC \Rightarrow 4 \cdot \frac{1}{2} |a \times b + b \times c + c \times a| = k \cdot \text{area of}$$

ΔABC

$$\Rightarrow 4 \text{ area of } \Delta ABC = k \cdot \text{area of } \Delta ABC \Rightarrow k = 4$$

22.(1) We have given that,

$$I = \int_{-\pi/4}^{3\pi/4} \frac{\sin x + \cos x}{e^{x-\pi/4} + 1} dx \quad \dots (i)$$

$$I = \int_{-\pi/4}^{3\pi/4} \frac{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}{(e^{\pi/4-x} + 1)} dx$$

Using identity, $\int_a^b f(x) dx = \int_b^a f(a+b-x) dx$

$$= \int_{-\pi/4}^{3\pi/4} \frac{\cos x + \sin x}{e^{\pi/4-x} (1 + e^{x-\pi/4})} dx$$

$$I = \int_{-\pi/4}^{3\pi/4} \frac{\cos x + \sin x}{e^{\pi/4-x} (1 + e^{x-\pi/4})} dx$$

$$I = \int_{-\pi/4}^{3\pi/4} \frac{\cos x + \sin x}{(1 + e^{x-\pi/4})} e^{x-\pi/4} dx \quad \dots (ii)$$

Adding equations (i) and (ii)

$$2I = \int_{-\pi/4}^{3\pi/4} \frac{(\cos x + \sin x) \left(1 + e^{x-\frac{\pi}{4}} \right)}{(1 + e^{x-\pi/4})} dx; \quad \int_{-\pi/4}^{3\pi/4} (\cos x + \sin x) dx$$

23.(0.5)

Consider, the given expression

$$\begin{aligned} & \lim_{x \rightarrow \infty} \sqrt{3x^2 + \sqrt{3x^2 + \sqrt{3x^2}}} - \sqrt{3x^2} \\ &= \lim_{x \rightarrow \infty} \frac{3x^2 + \sqrt{3x^2 + \sqrt{3x^2}} - 3x^2}{\sqrt{3x^2 + \sqrt{3x^2 + \sqrt{3x^2}}} + \sqrt{3x^2}} \quad \{\text{By rationalisation}\} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{3 + \sqrt{\frac{3}{x^2}}}}{\sqrt{3 + \sqrt{\frac{3}{x^2}}} + \sqrt{\frac{3}{x^2}}} = \frac{\sqrt{3}}{\sqrt{3} + \sqrt{3}} = \frac{1}{2} = 0.5 \end{aligned}$$

24.(1.13)

On solving the given equation after differentiating, we get :

$$f(x) = 3 \cos x - 2 \cos^2 x = -\left[\cos x - \frac{3}{4}\right]^2 + \frac{9}{8}; \quad f(x)_{\max} = \frac{9}{8} = 1.125 = 1.13$$

25.(56)

$$\begin{aligned} f(1) = 0 \text{ and } f(2) = 30 &\Rightarrow \int_0^2 f(x) dx + \int_0^{30} f^{-1}(y) dy = \int_0^1 f(x) dx + \int_1^2 f(x) dx + \int_1^2 tf'(t) dt \\ &= \left(x^4 + x^2 - 6x\right)_0^1 + \int_1^2 f(x) dx + \left(tf(t)\right)_1^2 - \int_1^2 f(t) dt = 56 \end{aligned}$$

26.(5)

$$1 + z + z^2 + \dots + z^{100} = (z - \alpha_1)(z - \alpha_2)(z - \alpha_3) \dots (z - \alpha_{100})$$

$$\text{Differentiating } \frac{1 + 2z + 3z^2 + \dots + 100z^{99}}{1 + z + z^2 + \dots + z^{100}} = \frac{1}{z - \alpha_1} + \frac{1}{z - \alpha_2} + \dots + \frac{1}{z - \alpha_{100}}$$

Put $z = 1$

$$\frac{1 + 2 + 3 + \dots + 100}{1 + 1 + 1 + \dots + 1} = \frac{1}{1 - \alpha_1} + \frac{1}{1 - \alpha_2} + \dots + \frac{1}{1 - \alpha_{100}}$$

$$\frac{100 \times 101}{2 \times 101} = -\left(\frac{1}{\alpha_1 - 1} + \frac{1}{\alpha_2 - 1} + \dots + \frac{1}{\alpha_{100} - 1}\right)$$

$$-50 = \frac{1}{\alpha_1 - 1} + \frac{1}{\alpha_2 - 1} + \dots + \frac{1}{\alpha_{100} - 1}$$

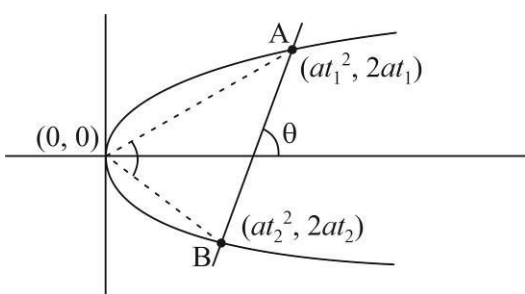
$$\sum_{r=1}^{100} \frac{1}{\alpha_r - 1} = -50 = 10\lambda; \quad \lambda = -5 \text{ so } |\lambda| = 5$$

Alternate :

$$\alpha_1, \alpha_2, \alpha_{100} \text{ are roots of } \sum_{k=0}^{100} z^k = 0$$

$$\begin{aligned} \text{Let } y &= \frac{1}{z-1} \Rightarrow z = 1 + \frac{1}{y} \\ \Rightarrow \frac{1}{\alpha_r - 1} \{r = 1; 100\} &\text{ are roots of } \sum_{k=0}^{100} \left(1 + \frac{1}{y}\right)^k = 0 \\ \Rightarrow \frac{\left(1 + \frac{1}{y}\right)^{101} - 1}{1/y} &= 0 \quad \Rightarrow \quad \frac{y[(y+1)^{101} - y^{101}]}{y^{101}} = 0 \\ \Rightarrow 101y^{100} + {}^{101}C_2 y^{99} + \dots &= 0 \quad \Rightarrow \quad \sum_{r=1}^{100} \frac{1}{\alpha_r - 1} = -\frac{{}^{101}C_2}{101} = -50 \end{aligned}$$

27.(2)

$$\begin{aligned} y^2 &= 8x \\ \text{Now, } t_1 t_2 &= -4 \\ \text{And } t_2 &= -t_1 - \frac{2}{t_1} \\ \frac{-4}{t_1} &= -t_1 - \frac{2}{t_1} \\ \Rightarrow -t_1^2 &= -2 \\ \Rightarrow t_1^2 &= 2 \\ y^2 &= 4ax \\ \frac{dy}{dx} &= \frac{4ax}{2y}; \quad \frac{dy}{dx} = \frac{1}{t} \end{aligned}$$


$$\text{Slope of normal} = -t \text{ and } \tan^2 \theta = t_1^2 = 2$$

28.(81)

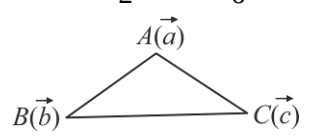
$$\begin{aligned} \text{Given } \frac{f(x)f'(x)}{\sqrt{1-(f(x))^4}} &\geq x \\ \frac{2f(x)f'(x)}{\sqrt{1-(f(x))^4}} &\geq x \end{aligned}$$

$$\text{Let } g(x) = \sin^{-1}(f(x))^2 - x^2 \text{ is non decreasing function}$$

$$\lim_{x \rightarrow a^+} g(x) \leq \lim_{x \rightarrow b^-} g(x)$$

$$\frac{\pi}{2} - a^2 \leq \frac{\pi}{6} - b^2; \quad a^2 - b^2 \geq \frac{\pi}{3}; \quad [a^2 - b^2] = 1$$

29.(3)

$$\begin{aligned} \vec{A}(\vec{a}) \quad \vec{B}(\vec{b}) \quad \vec{C}(\vec{c}) \\ \vec{a} + 2\vec{b} + 3\vec{c} = 0 \\ \text{Area of } \triangle ABC = \frac{1}{2} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}| \end{aligned}$$


$$\vec{a} \times \vec{b} = -\frac{3(\vec{a} \times \vec{c})}{2}$$

$$\vec{b} \times \vec{c} = -\frac{\vec{a} \times \vec{c}}{2}$$

$$\vec{c} \times \vec{a} = -\vec{a} \times \vec{c}$$

$$\text{Area of } \triangle ABC = \frac{1}{2} |-3(\vec{a} \times \vec{c})| = \frac{3}{2} |\vec{a} \times \vec{c}|$$

$$\triangle AOC = \frac{1}{2} |\vec{a} \times \vec{c}|$$

$$\frac{\text{arc } \triangle ABC}{\text{arc } \triangle AOC} = 3$$

30.(8) $E \rightarrow$ Die shows up 3.

$F \rightarrow$ Fair die picked

$U \rightarrow$ Unfair die picked

$$P(E) = P(F) P(E / F) + P(U) P(E / U) = \frac{1}{4} \times \frac{1}{6} + \frac{3}{4} \times \frac{2}{6}$$

$$P(F / E) = \frac{P(F \cap E)}{P(E)} = \frac{\frac{1}{4} \times \frac{1}{6}}{\frac{1}{4} + \frac{6}{24}} = \frac{1}{7}$$